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Endo-E2E-GS: End-to-end 3D reconstruction of endoscopic scenes using Gaussian Splatting☆

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ABSTRACT

Three-dimensional (3D) reconstruction is essential for enhancing spatial perception and geometric understanding in minimally invasive surgery. However, current methods like Neural Radiance Fields (NeRF) and 3D Gaussian Splatting (3DGS) often rely on offline preprocessing—such as COLMAP-based point clouds or multi-frame fusion—limiting their adaptability and clinical deployment. We propose Endo-E2E-GS, a fully end-to-end framework that reconstructs structured 3D Gaussian fields directly from a single stereo endoscopic image pair. The system integrates (1) a DilatedResNet-based stereo depth estimator for robust geometry inference in low-texture scenes, (2) a Gaussian attribute predictor that infers per-pixel rotation, scale, and opacity, and (3) a differentiable splatting renderer for 2D view supervision. Evaluated on the ENDONERF and SCARED datasets, Endo-E2E-GS achieves highly competitive performance, reaching PSNR values of 38.874/33.052 and SSIM scores of 0.978/0.863, respectively, surpassing recent state-of-the-art approaches. It requires no explicit scene initialization and demonstrates consistent performance across two representative endoscopic datasets. Code is available at: **https://github.com/Intelligent-Imaging-Center/Endo-E2E-GS**.

1. Introduction

3D reconstruction from endoscopic videos is a cornerstone technology in minimally invasive surgery (MIS), enhancing surgical navigation and operational precision by providing surgeons with augmented spatial perception of the surgical field. Beyond intraoperative guidance, this technology also supports crucial applications in virtual surgical simulation, medical training, and robotic automation [1–3]. In robotic-assisted procedures, precise 3D reconstruction is particularly vital for enabling real-time visual feedback and fine-grained control. However, endoscopic scenes present unique and significant challenges—such as sparse textures on tissue surfaces, a narrow field-of-view, frequent instrument occlusions, and non-rigid deformations—that limit the effectiveness of conventional 3D reconstruction methods [4–6].

To address these challenges, a variety of methods have been proposed. Traditional approaches include geometry-based stereo depth estimation [7–9], SLAM-based mapping [10–12], sparse warp field models [13,14], and voxel-based volumetric representations [15]. While these methods provide interpretable geometry, they struggle to accommodate non-rigid anatomical motion and often produce fragmented or visually discontinuous reconstructions in texture-scarce regions. Advances in neural rendering, particularly Neural Radiance Fields (NeRF) [16,17] have introduced a paradigm shift, encoding scenes into a continuous implicit function that markedly improved rendering quality and realism. Nevertheless, the significant computational demands and protracted per-scene training duration of NeRF, compounded by its lack of an explicit geometric representation, impede its practical application in time-sensitive surgical contexts [18].

As a powerful alternative, 3DGS [19] emerged, combining memory-efficient primitives with a differentiable rasterization pipeline. Compared to voxel grids, it avoids the need for dense volumetric storage;

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compared to NeRF, it significantly improves rendering speed and training efficiency, while retaining explicit geometric control. Despite this potential, current 3DGS-based methods still face multiple hurdles when applied to endoscopic scenes. Firstly, they typically rely on offline preprocessing steps, such as generating a sparse point cloud with COLMAP [20–22]; this process is unreliable in endoscopic videos with limited viewpoints and frequent occlusions. Secondly, most existing methods depend on per-scene optimization [23–25], a time-consuming process that fails to meet the real-time requirements of surgery. Finally, many feed-forward 3DGS frameworks [26–28] are designed for general object-level scenes and lack adaptability to the unique challenges of non-rigid tissue and poor textures found in MIS.

To this end, we propose **Endo-E2E-GS**, an end-to-end framework for structured 3D reconstruction from stereo endoscopic images that directly addresses the aforementioned limitations. Unlike existing 3DGS-based methods, Endo-E2E-GS reconstructs 3D Gaussian fields directly from a single stereo pair via a fully feed-forward architecture, eliminating any reliance on scene-specific preprocessing or offline optimization. The system integrates a stereo depth estimator and a Gaussian attribute predictor, both built upon DilatedResNet encoders. By systematically enlarging the receptive field, this design enhances both geometric and appearance feature extraction in texture-scarce surgical scenes, enabling robust, high-fidelity reconstruction towards real time performance. Our main contributions are summarized as follows:

- We propose Endo-E2E-GS, a fully end-to-end and feed-forward framework that reconstructs structured 3D Gaussian fields directly from a single stereo endoscopic image pair, without requiring any scene-specific geometry, point cloud initialization, or offline preprocessing.
- We extend conventional ResNet-based architectures by incorporating DilatedResNet into both the stereo depth estimation and Gaussian attribute prediction modules. This design enlarges the effective receptive field and enhances feature extraction in texture-scarce and anatomically complex surgical scenes.
- Our method achieves state-of-the-art rendering quality (PSNR: 38.874, SSIM: 0.978) on ENDONERF and SCARED, while supporting efficient 3D reconstruction suitable for surgical scenarios.

2. Related work

Implicit field 3D reconstruction has achieved significant progress in the past few years, with NeRF [16] becoming a foundational technique in photorealistic novel view synthesis. NeRF represents scenes via coordinate-based multilayer perceptrons (MLPs), mapping spatial locations and viewing directions to density and color. This paradigm enables continuous and high-fidelity 3D representations but lacks explicit surface geometry, limiting its downstream usability in robotics and clinical applications. To adapt NeRF for dynamic surgical scenes, several works have explored implicit modeling tailored for endoscopic data. EndoNeRF [4] leverages a dual-field representation inspired by D-NeRF [29], with separate networks modeling canonical space appearance and deformation fields, allowing it to account for non-rigid tissue motion. LerPlane [5] extends this idea by integrating a learnable plane representation into a temporal context, resulting in a 4D volume that markedly reduces training time while preserving temporal coherence. However, real-time rendering remains challenging due to the model's heavy dependency on optimization and scene-specific convergence. EndoSurf [6] shifts towards surface-centric modeling by replacing the density field with a signed distance function (SDF), providing explicit geometric constraints that improve reconstruction quality. Despite their innovations, all NeRF-based solutions remain fundamentally constrained by lengthy per-scene training, high computational demand, and a lack of direct geometric supervision—factors that render them unsuitable for time-sensitive and high-risk domains such as robotic surgery [18].

Gaussian Splatting-based 3D reconstruction offers an attractive alternative by introducing explicit 3D scene primitives with efficient, differentiable rasterization. Recent works have extended 3DGS [19] into the surgical domain, combining geometry modeling with dynamic motion estimation. EndoGS [20] adopts spatiotemporal importance optimization to prioritize visible regions and reduce redundancy in Gaussian rendering, addressing occlusion-heavy environments. Endo-Gaussian [21] and Endo4DGS [22] initialize Gaussians using DepthAnything estimations and subsequently perform scene-specific optimization to improve global coherence and appearance quality. SurgicalGS [23] proposes a hybrid initialization method using learned priors and motion masks, coupled with an additional regularization term to enhance geometric stability across frames. EndoSparse [25] introduces a knowledge distillation framework to handle sparse viewpoints, transferring geometric and appearance priors from large pretrained visual-language models. HFGS [24] contributes frequency-aware spatial and temporal modules, refining motion perception and spatial details by aligning high-frequency signals across time. LGS [30] tackles memory and efficiency bottlenecks by compressing Gaussian attributes and operating with low-resolution features, allowing scalable deployment in resource-constrained surgical setups. Deform3DGS [31] introduces motion-aware fusion and linear combination modeling to simulate tissue deformation during dynamic surgeries while producing denser initializations. Although these approaches demonstrate impressive performance in dynamic reconstruction, they universally depend on precomputed geometry (e.g., COLMAP), per-scene optimization, or offline depth fusion—making them incompatible with real-time, frame-level systems and less deployable in practical surgical settings.

Feed-forward 3D Gaussian Splatting methods aim to eliminate per-scene optimization entirely, enabling efficient inference by predicting Gaussian parameters directly from input images. These methods typically employ either monocular or stereo input and prioritize real-time performance, making them attractive for clinical deployment. Splatter Image [32] introduces an encoder–decoder architecture for single-view object reconstruction, predicting position, scale, and color for sparse Gaussians in a category-specific setting. PixelSplat [27] advances this line of work by incorporating an epipolar Transformer to reason about multi-view consistency and produce pixel-aligned probabilistic depth. While it improves correspondence across views, PixelSplat still suffers from noise amplification in depth maps and occasional geometric discontinuities. MVSplat [28] constructs a plane-sweeping cost volume to capture cross-view features and performs Gaussian prediction with learned view fusion. TranSplat [33] further incorporates depth-aware deformable attention and monocular priors to handle sparse camera viewpoints. GPS-Gaussian [26] utilizes stereo matching and cost aggregation to estimate Gaussian parameters in a generalizable, end-to-end manner, enabling dual-view inference. Despite their strengths, these approaches have been primarily developed for structured, object-centric environments and have not been validated in the complex setting of endoscopic surgery. Their generalizability remains uncertain in the presence of texture-scarce, specular, and non-rigid tissue conditions commonly found in clinical scenarios. This highlights the need for domain-specific solutions capable of real-time reconstruction without relying on scene-specific initialization or offline optimization.

3. Preliminaries

Depth Estimation serves as a fundamental component in 3D reconstruction pipelines, aiming to recover scene geometry from 2D image observations [34]. In this work, we adopt binocular stereo vision—a geometrically grounded approach—where depth estimation is achieved by computing disparity between rectified stereo image pairs [35]. The relationship between disparity and depth can be formulated using the perspective projection principle:

$$D = \frac{f \cdot b}{d}, \tag{1}$$

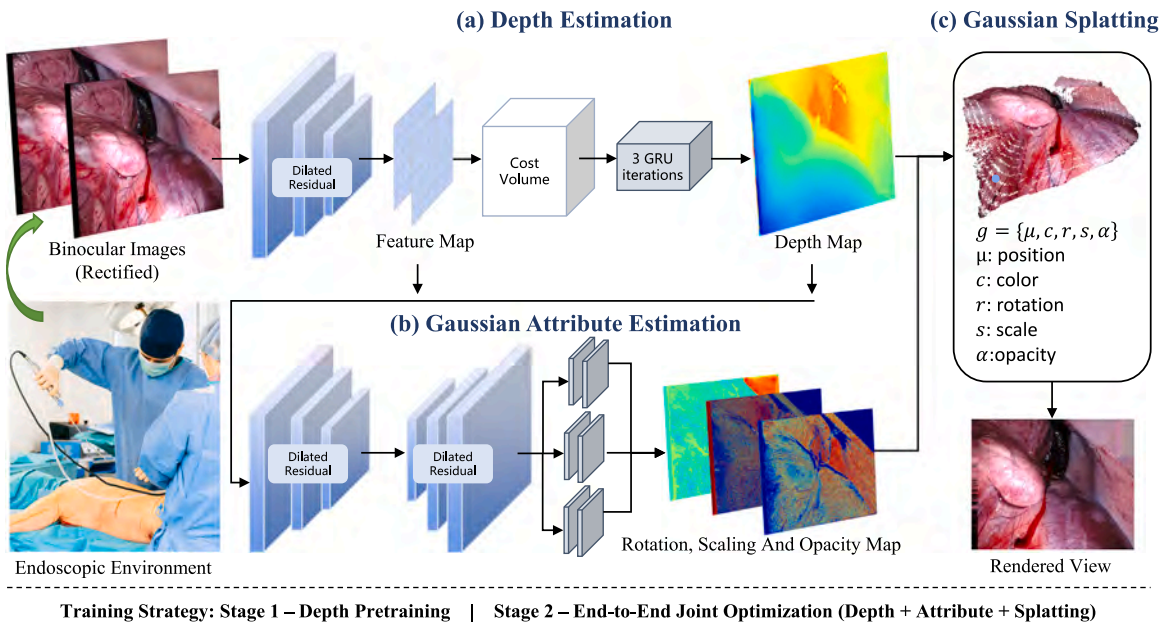


Fig. 1. Overview of the proposed Endo-E2E-GS framework. The system directly predicts a structured 3D Gaussian representation from a single stereo endoscopic image pair, without relying on scene-specific point clouds or multi-frame depth fusion. Stage 1 pretrains a depth estimation module with DilatedResNet and GRU-based cost volume refinement. Stage 2 jointly optimizes the Gaussian attribute prediction and Gaussian Splatting modules. The attribute module infers rotation, scale, and opacity from the depth and feature maps, forming a Gaussian field parameterized by $g = \{\mu, c, r, s, \alpha\}$. Differentiable rendering via Gaussian Splatting synthesizes 2D views for photometric supervision, enabling real-time, high-fidelity 3D reconstruction suitable for intraoperative applications.

where f is the focal length (in pixels), b is the baseline distance between stereo camera centers (in meters), d denotes pixel-level disparity, and D represents the estimated depth (in meters). This computed depth map serves as the geometric foundation for subsequent 3D representation and differentiable rendering.

3D Gaussian Splatting represents a scene using a collection of anisotropic 3D Gaussian primitives, each fully parameterized by its geometric and photometric attributes [19]. Specifically, each Gaussian is defined by its mean position $\mu \in \mathbb{R}^3$, RGB color $\mathbf{c} \in \mathbb{R}^3$, rotation quaternion $\mathbf{r} \in \mathbb{R}^4$, anisotropic scale $\mathbf{s} \in \mathbb{R}^3$, and opacity $\alpha \in [0, 1]$. The spatial distribution of the Gaussian is governed by a covariance matrix $\Sigma \in \mathbb{R}^{3\times3}$, derived from the rotation and scaling parameters as:

$$\Sigma = \mathbf{RSS}^\top\mathbf{R}^\top, \tag{2}$$

where $\mathbf{R}$ is the rotation matrix converted from quaternion $\mathbf{r}$, and $\mathbf{S} = \mathrm{diag}(\mathbf{s})$ is a diagonal scaling matrix. The density function of the Gaussian at a 3D point $\mathbf{x} \in \mathbb{R}^3$ is given by:

$$G(\mathbf{x}) = \exp\left(-\frac{1}{2}(\mathbf{x}-\mu)^\top\Sigma^{-1}(\mathbf{x}-\mu)\right). \tag{3}$$

During rasterization, each 3D Gaussian is projected onto the 2D image plane using differentiable splatting [36]. Given the camera projection matrix $\mathbf{W}$ and its Jacobian $\mathbf{J}$, the projected 2D mean μ' and covariance Σ' are computed as:

$$\mu' = \mathbf{W}\mu, \quad \Sigma' = \mathbf{JW}\Sigma\mathbf{W}^\top\mathbf{J}^\top. \tag{4}$$

The 2D Gaussian density function is then expressed as:

$$G'(\mathbf{x}) = \exp\left(-\frac{1}{2}(\mathbf{x}-\mu')^\top\Sigma'^{-1}(\mathbf{x}-\mu')\right). \tag{5}$$

The final rendered pixel color $\mathbf{C}(\mathbf{x})$ is computed via alpha compositing of all Gaussians along the viewing ray, following the over-operator blending rule [16]:

$$\mathbf{C}(\mathbf{x}) = \sum_{i=1}^{N}\mathbf{c}_i\alpha_i G'_i(\mathbf{x})\prod_{j=1}^{i-1}\left(1-\alpha_j G'_j(\mathbf{x})\right), \tag{6}$$

where N is the number of projected Gaussians sorted in front-to-back depth order.

All parameters $\{\mu, \mathbf{c}, \mathbf{r}, \mathbf{s}, \alpha\}$ are optimized end-to-end via gradient descent. The differentiability of the entire pipeline enables supervision through rendered images, encouraging photometric alignment and geometric consistency across the learned 3D Gaussian field.

4. Method

As illustrated in Fig. 1, the proposed Endo-E2E-GS framework reconstructs a structured 3D Gaussian field directly from a single stereo endoscopic image pair, without relying on any scene-specific initialization. The pipeline operates in two stages. In Stage 1, a binocular depth estimation network—comprising a DilatedResNet encoder and a Gated Recurrent Unit (GRU)-based cost volume refinement module—predicts a dense depth map from the stereo image pair (Section 4.1). In Stage 2, the predicted depth and extracted features serve as input to the Gaussian attribute estimation module, which outputs rotation, scale, and opacity parameters for each pixel (Section 4.2). These parameters, combined with color and position information, form a structured set of 3D Gaussians. Differentiable rendering via Gaussian Splatting enables end-to-end supervision through photometric loss, achieving real-time and high-fidelity reconstruction (Section 4.3).

4.1. Depth estimation

Depth estimation in binocular stereo vision seeks to infer 3D structure by identifying dense pixel correspondences between a rectified stereo image pair. The core principle involves calculating disparity—the horizontal displacement between matched pixels across views—which can be transformed into depth using known camera intrinsics and stereo baseline. Following a standard four-stage pipeline [37]—comprising feature extraction, cost volume construction, disparity regression, and post-refinement—our design incorporates a dilated residual encoder and a GRU-based refinement module tailored for endoscopic imagery with texture-scarce regions. Compared to standard ResNet encoders, DilatedResNet expands the receptive field with minimal overhead, enabling better context aggregation [38]. This helps

reduce stereo matching ambiguities in textureless or unevenly lit regions, which are common in endoscopic scenes. The feature extractor is based on a DilatedResNet that processes input images $I_L, I_R \in \mathbb{R}^{H\times W\times 3}$ through an initial 5 × 5 convolutional layer (stride=1, padding=2), followed by three residual groups with channel dimensions of 64, 96, and 128, respectively. Each group contains two residual blocks composed of 3 × 3 dilated convolutions (dilation rate=2, stride=2), group normalization, and ReLU activation. The encoder downsamples the input by a factor of 8, resulting in deep feature maps $F_L, F_R \in \mathbb{R}^{H/8\times W/8\times 128}$.

The extraction is defined as:

$$F_L = \mathcal{D}(I_L;\theta_d), \quad F_R = \mathcal{D}(I_R;\theta_d), \tag{7}$$

where $\mathcal{D}$ denotes the DilatedResNet with parameters θ_d shared across both views.

A cost volume is constructed by computing feature similarity across a set of discrete disparity levels:

$$C(u,v,d) = F_L(u,v)^\top F_R(u-d,v), \tag{8}$$

where (u,v) denotes spatial location and d the disparity hypothesis in the range $[0, D_{max})$.

To regress disparity, we adopt a ConvGRU-based refinement module [39]. At each time step, the disparity estimate is iteratively updated based on the cost volume:

$$d_{t+1} = \Gamma\left(d_t, C(u,v,d_t)\right), \quad t \in \{0,1,2\}, \tag{9}$$

where Γ represents a convolutional GRU consisting of three 3 × 3 convolutional gates (reset, update, and candidate) applied to the concatenation of hidden and input states. This iterative scheme enables context-aware refinement without requiring costly 3D convolutions.

Finally, the refined disparity map is converted into metric depth using the perspective projection equation (Eq. (1)). The resulting depth map D serves as the positional prior for subsequent 3D Gaussian parameter prediction.

4.2. Gaussian attribute estimation

For every pixel in the rectified left image, we generate a corresponding 3D Gaussian, resulting in a dense and structured Gaussian field that aligns with the image grid. Each 3D Gaussian is parameterized by a set $g = \{\mu, c, r, s, \alpha\}$, where $\mu \in \mathbb{R}^3$ denotes the 3D position of the Gaussian center, $c \in \mathbb{R}^3$ is its color in RGB space, $r \in \mathbb{R}^4$ represents its rotation as a unit quaternion, $s \in \mathbb{R}^3$ denotes anisotropic scale along each axis, and $\alpha \in [0,1]$ is the opacity coefficient used for volumetric blending. These attributes jointly determine the geometric and photometric behavior of each surface element: μ anchors the Gaussian in 3D space, c captures local appearance under the Lambertian reflectance assumption, s defines its spatial support, r aligns the Gaussian with the underlying surface orientation, and α reflects local visibility or confidence. Together, they support accurate, geometry-aware rendering via Gaussian splatting, ensuring geometric and photometric consistency across views.

The 3D position μ is computed by reprojecting each pixel (u,v) using the predicted depth D from Section 4.1, combined with the intrinsic matrix K and stereo extrinsics. Color c is directly sampled from the original RGB input, similar to GPS-Gaussian [26], under the assumption of Lambertian reflectance, which is reasonable for most endoscopic tissue surfaces. This design avoids the need for modeling complex illumination or reflectance fields, while providing sufficiently realistic appearance.

To estimate the remaining parameters r, s, and α, we construct an encoder–decoder architecture that jointly encodes depth and appearance cues. Specifically, a depth encoder based on DilatedResNet is applied to the predicted depth map, while a ResNet-18 backbone extracts RGB features. These multi-scale features are fused hierarchically across three residual decoding blocks. At each decoding stage, RGB and depth features are concatenated and processed by two residual blocks composed of 3 × 3 convolutions, Group Normalization, and ReLU activation. The decoding proceeds from coarse (1/8 resolution) to fine (full resolution), allowing rich geometry-aware information to flow into the output parameter maps.

The final decoder features are processed by three separate convolutional heads to predict rotation, scale, and opacity maps. The rotation output is normalized to form a unit quaternion, which provides a continuous representation of 3D rotations. Unlike Euler angles that suffer from gimbal lock when two axes align and a degree of freedom is lost, quaternions maintain full rotational freedom throughout the space. The scale parameters are activated through a softplus function [40] to maintain positive values, preventing numerical instability and degenerate Gaussians. Opacity is constrained to the interval $[0,1]$ using a sigmoid function, which facilitates smooth alpha blending during splatting-based rendering. These post-processing operations guarantee that the estimated parameters remain numerically stable and physically plausible throughout training and inference.

4.3. End-to-end optimization

The structured 3D Gaussian field, parameterized as described in Section 4.2, is rendered into 2D images using differentiable Gaussian Splatting [19], following the projection and compositing process detailed in Section 3. This rendering pipeline enables view synthesis from arbitrary viewpoints by simulating the projection and accumulation of anisotropic Gaussian densities along the viewing ray. As a result, photometric feedback can be directly applied for end-to-end supervision of both geometric and appearance parameters.

To train the full system, we formulate a multi-objective loss function that jointly supervises depth estimation and 3D rendering. The overall objective consists of two components: a depth regression loss and a photometric reconstruction loss, weighted by coefficients w_1 and w_2, respectively. For depth supervision, we adopt the L1 loss to penalize the absolute error between predicted depth D and ground truth $\hat{D}$, promoting pixel-wise depth accuracy. For image-level supervision, we combine the Structural Similarity Index (SSIM) and the Mean Absolute Error (MAE) to jointly enforce perceptual fidelity and photometric alignment between the synthesized image $\hat{I}$ and the reference view I.

The total loss is defined as:

$$L_{\text{total}} = w_1 L_{\text{depth}} + w_2 L_{\text{recon}}, \tag{10}$$

where

$$L_{\text{depth}} = \frac{1}{N}\sum_{i=1}^{N}\left|D_i - \hat{D}_i\right|, \tag{11}$$

and

$$L_{\text{recon}} = \alpha\cdot(1-\text{SSIM}(I,\hat{I})) + \beta\cdot\text{MAE}(I,\hat{I}). \tag{12}$$

In our experiments, we set $w_1 = 0.6$ and $w_2 = 0.4$ based on ablation results (see Fig. 4), which indicate that assigning slightly more emphasis to depth estimation improves convergence and final reconstruction quality. For the photometric loss, we set $\alpha = 0.8$ and $\beta = 0.2$ to balance structural preservation and intensity-level accuracy. This combination encourages the network to produce high-fidelity geometry while maintaining realistic visual appearance, thereby supporting robust 3D reconstruction suitable for surgical scenarios. To further stabilize depth estimation, we apply segmentation masks to exclude occluded regions and surgical tools from the depth supervision, which helps reduce invalid correspondences and improves disparity learning under occlusions. However, naively applying masks to the rendering loss reduces the amount of valid photometric supervision, leading to incomplete reconstructions in masked areas, as analyzed in our ablation study (Section 5.4).

5. Experiments

5.1. Datasets and metrics

We evaluate our method on two publicly available stereo endoscopic datasets: SCARED [41] and ENDONERF [4]. **SCARED** is a large-scale abdominal dataset with over 23,000 stereo RGBD image pairs at 1280 × 1024 resolution, collected from nine *in vivo* porcine subjects using the Da Vinci Xi system. Data from each subject forms a distinct subset, each containing four to five keyframe sequences from different intraoperative configurations. **ENDONERF** focuses on prostatectomy scenarios and contains 219 stereo pairs at 640 × 512 resolution, captured using a Da Vinci surgical robot from a single viewpoint. It includes stereo-derived depth maps and manually annotated tool masks, with scenes exhibiting substantial non-rigid tissue motion.

We assess performance using a comprehensive set of metrics for both rendering quality and geometric accuracy.

Rendering Quality. We evaluate image fidelity using three standard metrics:

- **Peak Signal-to-Noise Ratio (PSNR)** measures pixel-level reconstruction fidelity based on the Mean Squared Error (MSE), where higher is better.

$$\mathrm{MSE}(I,\hat{I}) = \frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}\left(I(i,j)-\hat{I}(i,j)\right)^2, \tag{13}$$

$$\mathrm{PSNR} = 10\cdot\log_{10}\left(\frac{\mathrm{MAX}_I^2}{\mathrm{MSE}}\right) \tag{14}$$

- **Structural Similarity Index (SSIM)** assesses perceptual similarity based on luminance, contrast, and structure.

$$\mathrm{SSIM}(I,\hat{I}) = \frac{(2\mu_I\mu_{\hat{I}}+C_1)(2\sigma_{I\hat{I}}+C_2)}{(\mu_I^2+\mu_{\hat{I}}^2+C_1)(\sigma_I^2+\sigma_{\hat{I}}^2+C_2)}, \tag{15}$$

  where μ and σ denote the local mean and standard deviation, respectively, and $\sigma_{I\hat{I}}$ is the local covariance.
- **Learned Perceptual Image Patch Similarity (LPIPS)** measures perceptual distance using deep features.

$$\mathrm{LPIPS} = \sum_l \frac{1}{H_l W_l}\sum_{h,w}\left\|w_l \odot \left(\phi_l(I)_{h,w}-\phi_l(\hat{I})_{h,w}\right)\right\|_2^2, \tag{16}$$

  where ϕ_l denotes the lth layer feature map from a pretrained network, and w_l is a learned channel-wise weight.

Geometric Accuracy. We evaluate the recovered 3D structure using four metrics:

- **End-Point Error (EPE)** computes the average absolute error between predicted ($\hat{D}$) and ground-truth (D) depth maps.

$$\mathrm{EPE} = \frac{1}{N}\sum_{i=1}^{N}|D_i-\hat{D}_i|. \tag{17}$$

- **1-Pixel Accuracy (1pix)** is the percentage of pixels whose EPE is less than 1 pixel.

$$1\mathrm{pix} = \frac{100}{N}\sum_{i=1}^{N}\mathbf{1}_{\{|D_i-\hat{D}_i|<1\}}, \tag{18}$$

  where $\mathbf{1}_{\{\cdot\}}$ is the indicator function.
- **Chamfer Distance (CD)** measures the average bidirectional distance between predicted (P) and ground-truth (Q) point clouds.

$$\mathrm{CD}(P,Q) = \frac{1}{|P|}\sum_{p\in P}\min_{q\in Q}\|p-q\|_2^2 + \frac{1}{|Q|}\sum_{q\in Q}\min_{p\in P}\|q-p\|_2^2. \tag{19}$$

- **Normal Consistency (NC)** evaluates the angular similarity between predicted ($\hat{n}$) and ground-truth (n) surface normals.

$$\mathrm{NC} = \frac{1}{N}\sum_{i=1}^{N}\left|n_i^{\top}\hat{n}_i\right|. \tag{20}$$

Efficiency. Runtime performance is reported in frames per second (FPS).

5.2. Implementation details

Training Procedure. Our framework is trained in two sequential stages to ensure both geometric accuracy and photorealistic quality. First, the depth estimation module is pretrained independently for 60,000 iterations using a supervised L1 loss. This stage focuses on stabilizing depth predictions, especially in texture-scarce endoscopic regions. Second, the entire Endo-E2E-GS pipeline is jointly optimized for 100,000 iterations. This stage uses the combined loss function defined in Section 4.3, with weights for the depth and reconstruction terms set to $w_1 = 0.6$ and $w_2 = 0.4$, respectively, as determined by our ablation study (Fig. 4).

Hyperparameters and Optimization. Across both stages, we use the AdamW optimizer [42] with parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of 1×10^{-5}. The learning rate is initialized to 2×10^{-4} and decayed following a cosine annealing schedule. We use a batch size of 6 for the pretraining stage and 2 for the joint optimization stage to accommodate the memory requirements of the full rendering pipeline. To improve computational efficiency, we enable mixed precision training via PyTorch's Automatic Mixed Precision (AMP).

Evaluation Protocol. Our evaluation is designed to rigorously assess model robustness and generalization. We train our model exclusively on the `dataset3/keyframe1` subset of the SCARED dataset. We then perform two types of testing: (1) To evaluate robustness, we test the trained model on the remaining unseen subsets of SCARED, assessing its performance across different subjects and intraoperative conditions. (2) To evaluate generalization, we test the same model on the entire ENDONERF dataset—a different anatomical domain—without any fine-tuning. For fair comparison, all metrics are computed on the foreground region, excluding areas occluded by surgical tools using the provided masks.

Environment and Assumptions. All experiments are conducted on a single NVIDIA RTX A5000 GPU using PyTorch and CUDA 11.7. The datasets used in this work provide rectified stereo pairs along with intrinsic and extrinsic calibration parameters. As a result, no additional preprocessing—such as denoising, photometric enhancement, or rectification—is required. The rendering module is entirely implemented in PyTorch without any gradient interruption operations, ensuring that the gradients from the rendering loss can be smoothly backpropagated to the depth branch. The depth estimation module contains 3.9 million parameters, while the full Endo-E2E-GS pipeline consists of 5.14 million parameters in total.

5.3. Performance comparisons

We compare Endo-E2E-GS against four state-of-the-art reconstruction methods for endoscopic scenes, spanning both NeRF-based and 3D Gaussian Splatting-based frameworks. Among them, EndoSurf [6] is built upon neural radiance fields, while EndoGaussian [21], Endo4DGS [22], and Deform3DGS [31] leverage 3D Gaussian primitives. Pretrained weights are used when available (e.g., Endo4DGS), and other baselines are retrained using their original configurations for fair comparison.

Quantitative results are reported in Table 1. Our method achieves the best performance across all three metrics—PSNR, SSIM, and LPIPS—on both ENDONERF and SCARED datasets. In particular, Endo-E2E-GS improves PSNR by 0.53 dB over the next-best method on

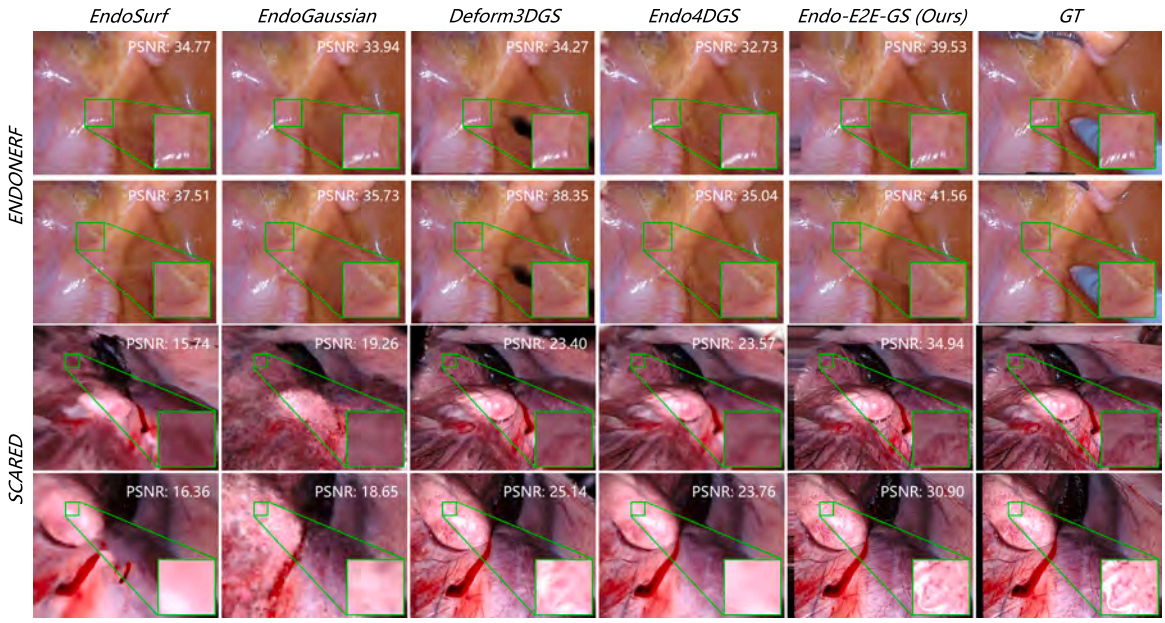


Fig. 2. Qualitative comparisons on the ENDONERF [4] and SCARED [41] datasets. Each row shows reconstruction results from different methods. Zoom-in regions were selected specifically to illustrate structural details, texture sharpness, and handling of specular highlights. Compared to NeRF-based (EndoSurf) and 3D Gaussian Splatting-based baselines (EndoGaussian, Deform3DGS, Endo4DGS), our Endo-E2E-GS generally preserves sharper textures and yields higher PSNR values across the evaluated scenes. Ground truth (GT) views are included for reference.

Table 1
Quantitative comparison with state-of-the-art methods on ENDONERF and SCARED datasets, chosen due to their representative use of NeRF-based and 3D Gaussian Splatting-based techniques. Endo-E2E-GS achieves superior reconstruction quality without requiring scene-specific initialization.

Method	ENDONERF (640 × 512)			SCARED (1280 × 1024)					Preprocessing
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	FPS	GPU Memory	
EndoSurf [6]	35.011	0.956	0.120	14.794	0.599	0.461	0.05	17GB	COLMAP
EndoGaussian [21]	37.316	0.958	0.069	19.253	0.621	0.491	147	2GB	COLMAP
Endo4DGS [22]	36.225	0.946	0.064	22.677	0.672	0.418	73	4GB	COLMAP
Deform3DGS [31]	38.343	0.961	0.063	30.112	0.861	0.165	210	1GB	PointCloud
Endo-E2E-GS (ours)	**38.874**	**0.978**	**0.054**	**33.052**	**0.863**	**0.152**	11	8GB	**None**

ENDONERF, and by 2.94 dB on SCARED, reflecting its superior reconstruction quality. These results highlight the effectiveness of our depth-guided design in preserving fine geometric structure and photometric consistency. The current pipeline achieves 11 FPS on the SCARED dataset (1280 × 1024 resolution), with depth estimation at 12 FPS and rendering at 95 FPS. While not yet real-time, Endo-E2E-GS reduces external dependencies and memory usage (8 GB), indicating its suitability for further optimization towards practical deployment in surgical environments.

Fig. 2 provides a visual comparison of rendering results across representative frames from both ENDONERF and SCARED datasets. Endo-E2E-GS consistently delivers the most faithful reconstructions, with sharper textures, better preservation of tissue boundaries, and improved recovery of overexposed regions. In the ENDONERF examples (top two rows), our method more accurately reproduces high-frequency anatomical details and specular reflections. In contrast, baselines such as EndoSurf and Endo4DGS produce blurry textures or lose contrast in saturated areas. On the more complex and high-resolution SCARED dataset (bottom two rows), Endo-E2E-GS shows clear advantages in regions with strong illumination, tool-induced shadows, and deformable tissues. The zoomed-in patches emphasize its ability to maintain structural coherence where other methods exhibit severe smoothing or overblending. These observations are consistent with the quantitative gains shown in Table 1.

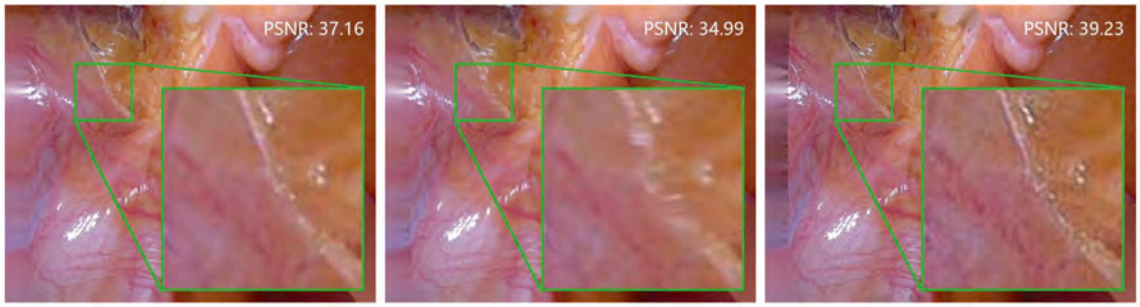
Table 2
Quantitative ablation study on the SCARED dataset. We evaluate rendering quality using PSNR, SSIM, and LPIPS; depth accuracy using end-point error (EPE) and 1-pixel accuracy ratio; and 3D geometric fidelity is evaluated using Chamfer Distance (CD) and Normal Consistency (NC). Lower CD and higher NC indicate better 3D shape preservation.

Method	Rendering			Geometry (Depth + 3D)			
	PSNR↑	SSIM↑	LPIPS↓	EPE↓	1 pix↑	CD↓	NC↑
Full model	**38.874**	**0.978**	**0.054**	**0.358**	**0.891**	**0.018**	**0.904**
w/o DilatedResNet	37.233	0.947	0.061	0.397	0.886	0.025	0.875
w/o Pretrain Depth	36.467	0.925	0.072	0.419	0.883	0.031	0.851

5.4. Ablation studies

(1) Impact of Encoder Architecture and Depth Pretraining. We first evaluate the impact of the DilatedResNet-based encoder and depth pretraining. In the "w/o DilatedResNet" variant, we replace the encoder with a standard ResNet-18 to assess the effect of receptive field enlargement. In the "w/o Pretraining" variant, the model is trained end-to-end from scratch without using any pretrained weights for the depth estimator. The visual and quantitative results are shown in Fig. 3 and Table 2, respectively.

Quantitatively, the full model demonstrates improved performance compared to the evaluated variants. Removing the DilatedResNet leads to a 1.64 dB drop in PSNR and a 0.039 increase in depth error (EPE),



(a) w/o DilatedResNet (b) w/o Pretrain Depth (c) Full Model

Fig. 3. Qualitative ablation on the "Pulling Soft Tissues" scene from the ENDONERF dataset, highlighting the necessity of DilatedResNet and depth pretraining in preserving fine structures and reducing geometric distortions. (a) Without DilatedResNet, reconstructions lack fine structure; (b) Without depth pretraining (i.e., training from scratch), distortions occur near deformable tissue boundaries.

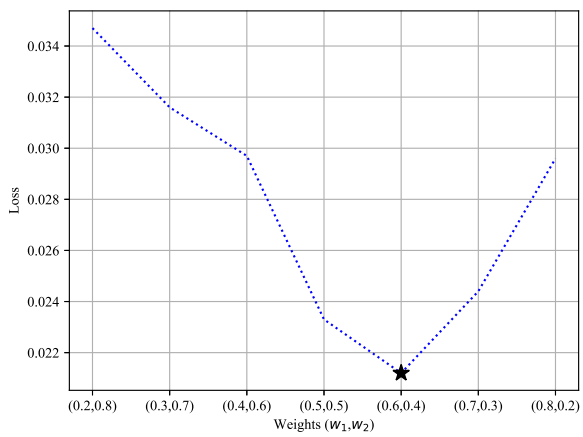


Fig. 4. Losses for different weight combinations of w_1, w_2. The one with an asterisk mark is selected for the subsequent experiments.

while skipping depth pretraining causes a further 2.41 dB drop and 0.061 increase in EPE. The 1-pixel accuracy also decreases accordingly. These results suggest that depth supervision plays a more critical role than encoder architecture alone, underscoring the significance of geometric priors in guiding 3D Gaussian Splatting models. To further assess 3D structural fidelity, we report Chamfer Distance (CD) and Normal Consistency (NC) in Table 2. These metrics are computed by back-projecting predicted and ground-truth depth maps into 3D point clouds using camera intrinsics. Ablation results show that removing DilatedResNet or depth pretraining noticeably degrades CD and NC, underscoring their importance for accurate shape reconstruction.

As illustrated in Fig. 3, removing the DilatedResNet leads to blurred reconstructions with degraded texture detail, while omitting depth pretraining results in localized geometric distortions, particularly in non-rigid regions. These artifacts highlight the importance of both components in learning stable spatial representations.

(2) Loss Weight Sensitivity. We investigate the effect of loss balancing by varying the weights w_1 and w_2 for the depth estimation and refinement stages. As shown in Fig. 4, the lowest total loss is achieved when $w_1 = 0.6$ and $w_2 = 0.4$, suggesting that the initial depth estimation has slightly more influence on convergence than the refinement stage. Larger imbalances between the two weights result in unstable training or suboptimal depth maps.

(3) Learning-Based vs. Traditional Depth Estimation. To evaluate the necessity of a learned depth estimator, we replace our network with the classical Semi-Global Block Matching (SGBM) algorithm from OpenCV and retain the rest of the pipeline. As shown in Table 3, SGBM produces significantly worse PSNR, SSIM, LPIPS, and EPE, indicating degraded rendering quality and depth accuracy. In contrast, our supervised model generates sharper, more reliable depth maps, especially in low-texture and occluded regions common in surgical scenes.

Table 3
Photometric and depth accuracy comparison between our method and SGBM on the SCARED dataset.

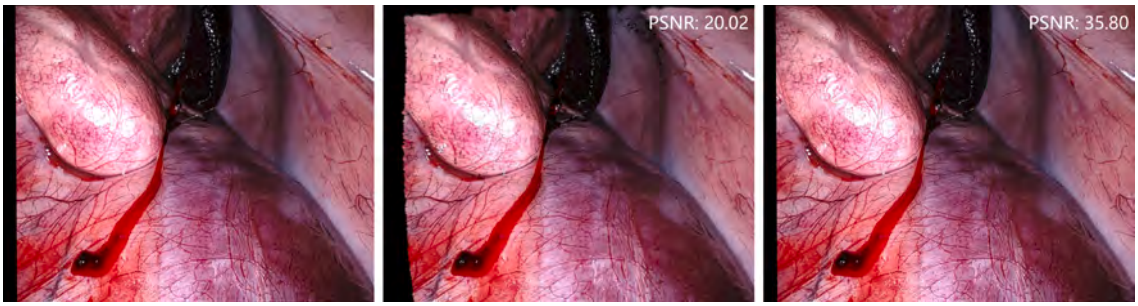
Depth Method	PSNR↑	SSIM↑	LPIPS↓	EPE↓	1 pix↑
SGBM (OpenCV)	32.76	0.712	0.093	0.578	0.620
Ours (Supervised)	**38.87**	**0.978**	**0.054**	**0.358**	**0.891**

(4) Effect of Surgical Tool Masks. We further investigate the impact of surgical tool masks on reconstruction quality. As shown in Fig. 5, applying masks during training lowers the PSNR from 35.80 to 20.02 and leads to incomplete reconstructions in the masked regions. This degradation arises because excluding masked pixels from the rendering loss reduces the amount of valid photometric supervision. In contrast, training without masks preserves more image content and achieves higher quantitative scores. These results indicate that while masks can improve depth estimation by removing invalid correspondences, their direct use in the rendering stage weakens photometric supervision and lowers overall visual fidelity.

5.5. Visualization of learned Gaussian attributes

To interpret the internal behavior of the Gaussian parameter estimation module, we visualize the predicted opacity and scaling maps in Fig. 6. The opacity map highlights the model's ability to suppress unreliable or outlier points by assigning them near-zero transparency. These points—typically located at the edges of occlusions or in texture-deficient regions—are rendered in blue and effectively ignored in the compositing process. In contrast, high-opacity regions in red or yellow concentrate on structurally salient tissue areas, ensuring they dominate the final appearance.

The scaling map further reveals how the model adaptively shapes each Gaussian based on local geometry and depth. Warmer regions, which represent Gaussians with larger scales, frequently appear in distant areas or low-texture surfaces where precision is harder to maintain. This design allows the model to maintain scene coverage while reducing overfitting to noisy details. Together, these maps confirm that the learned parameters not only improve rendering fidelity but also contribute to model robustness by regulating uncertainty and spatial resolution dynamically. We do not visualize rotation parameters, as they are encoded as 4D quaternions and lack a spatially interpretable scalar representation.



(a) Input Left Image (b) With Mask Render (c) w/o Mask Render

Fig. 5. Ablation on tool masks. (a) Input left image, (b) rendering result with mask, and (c) rendering result without mask. Using masks reduces photometric supervision and lowers fidelity (20.02 PSNR), whereas training without masks preserves higher fidelity (35.80 PSNR).

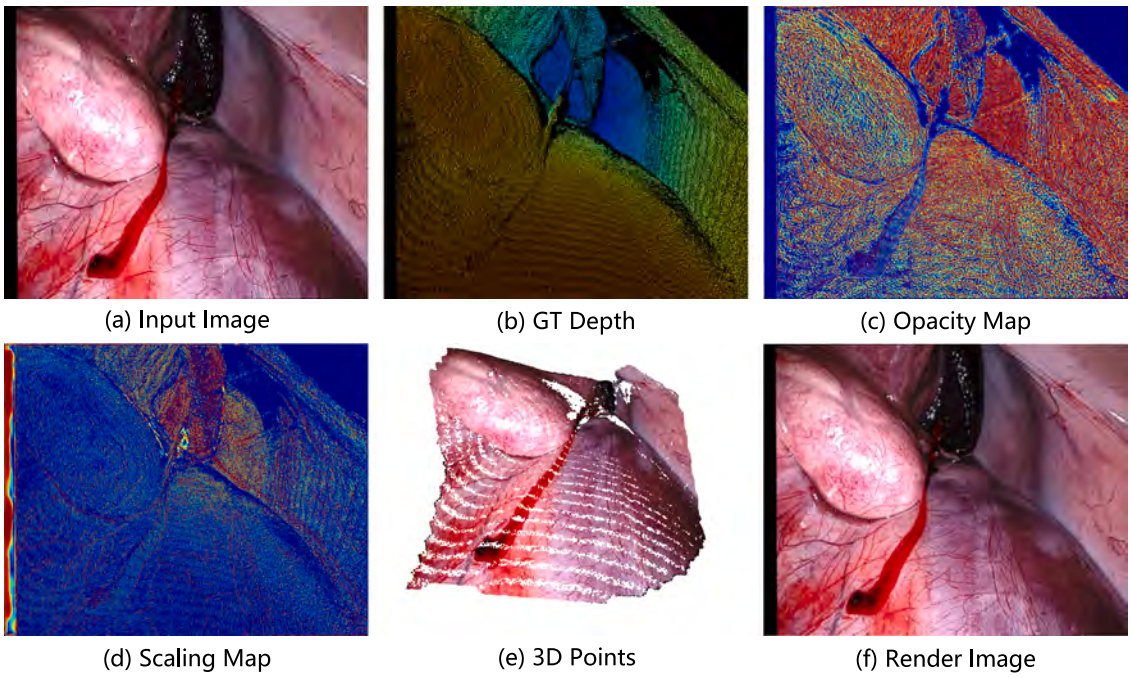


Fig. 6. Visualization of learned Gaussian attributes. (a) Input image; (b) ground-truth depth; (c) predicted opacity map, where low-opacity regions (blue) correspond to filtered outliers; (d) scaling map, where warmer colors indicate larger Gaussians often associated with distant surfaces; (e) reconstructed 3D Gaussian points; (f) final rendered image. The model adaptively suppresses uncertain regions and adjusts scale based on depth and geometry.

5.6. Robustness under challenging conditions

To evaluate the robustness of Endo-E2E-GS in complex intraoperative scenarios, we conduct additional testing on previously unseen subsets of the SCARED dataset that include diverse challenges such as specular reflections, strong lighting variation, tool occlusions, and active bleeding. These subsets were not used during training, enabling a realistic assessment of the model's performance in challenging cases.

As demonstrated in our qualitative evaluation (Fig. 7), Endo-E2E-GS exhibits remarkable robustness to several of the most common visual challenges in surgery. For instance, in scenes with active bleeding, the model maintains high-fidelity geometric and photorealistic reconstruction, correctly handling the significant variation in color and texture (fourth row). Similarly, when faced with uneven lighting and dark peripheral regions, the model successfully preserves the underlying geometry, even if the rendered view shows some intensity inconsistencies (second row). This indicates that the core depth estimation is resilient to moderate appearance changes.

The performance boundaries of our method are reached in extreme scenarios where the fundamental assumptions of stereo vision are violated. In cases of strong specular reflections (first row), the loss of valid pixel correspondences leads to depth estimation failure and rendering artifacts. Likewise, heavy tool-induced occlusions (third row) result in missing visual information, causing localized geometry loss.

It is important to note that our framework has inherent mechanisms to mitigate some of these issues. The opacity-aware rendering, for example, helps suppress artifacts from uncertain regions. This design also provides some intrinsic robustness to tool occlusions even without explicit masks. Nevertheless, fully conquering these extreme cases remains a key challenge for future work, which may involve incorporating reflection-robust loss functions or more sophisticated temporal modeling to fill in missing information.

6. Discussion

Our work demonstrates that high-fidelity, feed-forward 3D reconstruction for endoscopy is achievable. The success of Endo-E2E-GS stems from the synergy of its components: the DilatedResNet backbone effectively aggregates context in texture-scarce regions, depth pretraining stabilizes the geometric foundation, and the opacity-aware renderer robustly handles uncertainty from occlusions and specular highlights. These design choices are empirically validated by our ablation studies.

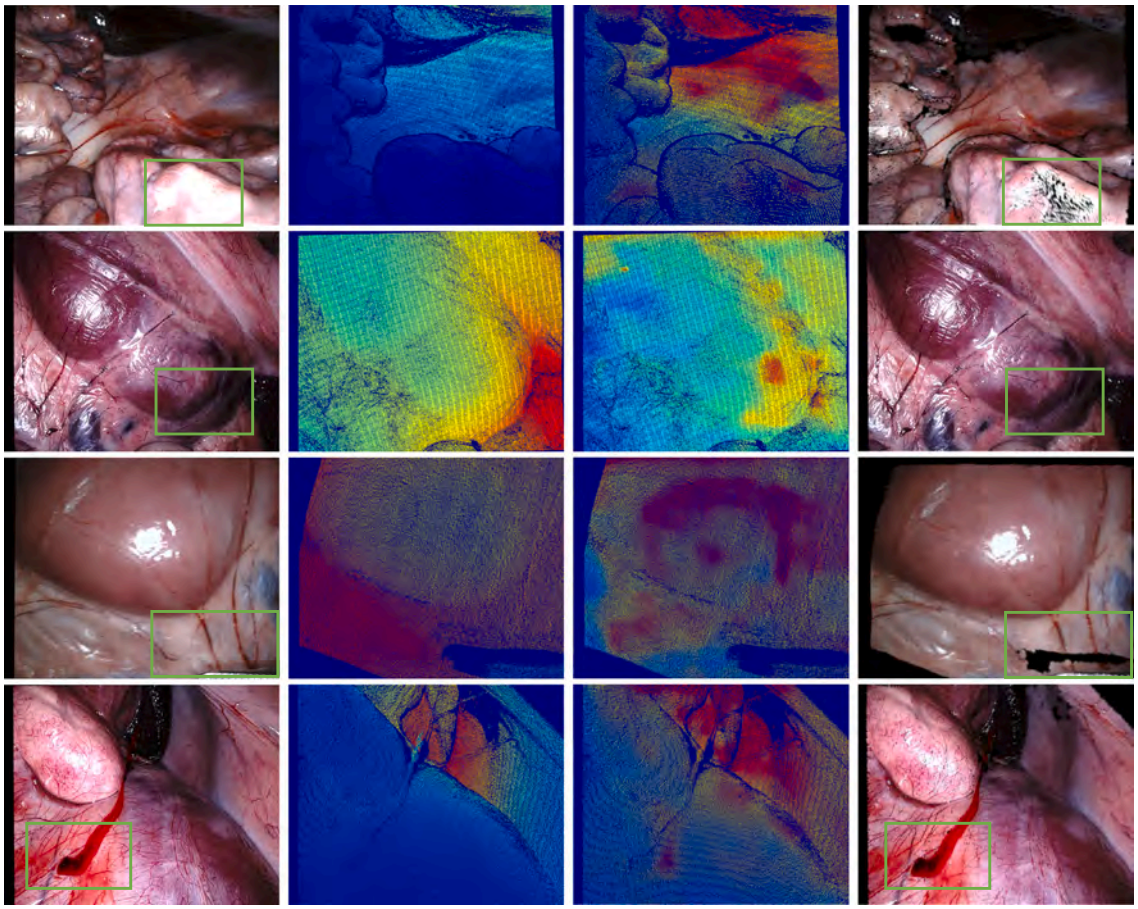


Fig. 7. Qualitative evaluation of Endo-E2E-GS on challenging samples from the SCARED dataset. Each row corresponds to a representative scenario: (1) strong specular reflections on moist tissues cause depth estimation failure and saturation artifacts in the rendering; (2) uneven lighting and dark peripheral regions introduce intensity inconsistencies; (3) tool-induced occlusions lead to geometry loss and spatial drift; (4) bleeding regions have minimal impact on depth prediction, demonstrating robustness to color and texture variation. From left to right: (a) input stereo left image, (b) ground-truth depth, (c) predicted depth, and (d) rendered view from the Gaussian field.

However, a critical analysis reveals key limitations that define the path towards clinical viability. While robust to moderate appearance changes, the model's performance degrades under strong specular reflections or heavy occlusions. This is primarily because depth estimation uncertainty propagates into the Gaussian field, creating geometric artifacts. This highlights a fundamental challenge for any method relying on purely passive stereo vision in such conditions.

Moreover, precomputed tool masks are employed only in the depth supervision stage to suppress invalid correspondences under occlusions. In contrast, the rendering stage is trained without masks to preserve photometric supervision. As shown in our ablation study (Section 5.4), applying masks in the rendering loss substantially lowers fidelity (PSNR drops from 35.80 to 20.02) due to the reduced amount of valid photometric supervision. This trade-off indicates that while masks are effective for stabilizing geometry learning, their direct use in rendering weakens supervision and lowers visual fidelity.

Furthermore, practical deployment faces three major hurdles. In terms of efficiency, our current pipeline runs at 11 FPS at 1280×1024 resolution, with depth estimation at 12 FPS and rendering at 95 FPS, requiring only 5.14M parameters and 8 GB GPU memory. These quantitative results highlight that the framework is compact and efficient, but still falls short of the $\geq$ 30 FPS typically required for surgical navigation. Second, the lack of temporal modeling means each frame is reconstructed independently, which could lead to a lack of coherence in live video streams. Third, although our experiments are limited to Da Vinci datasets, cross-dataset testing (training on SCARED and testing on ENDONERF) demonstrates a degree of generalization across anatomical domains and optical setups, but broader validation remains needed. Finally, we have not yet conducted system-level integration studies, such as synchronization with surgical tool tracking, latency evaluation in live operating conditions, or prototype deployment.

Addressing these issues forms a clear research trajectory. Future work must focus on three core areas: (1) Algorithmic Robustness, by developing uncertainty-aware filtering and reflection-robust loss functions; (2) Temporal Coherence and Efficiency, by integrating temporal priors or recurrent structures and exploring model compression; and (3) System-Level Integration and Generalization, by tackling challenges like latency, tool tracking, and domain adaptation for cross-platform deployment.

7. Conclusion

In this paper, we presented **Endo-E2E-GS**, a novel end-to-end framework for 3D reconstruction from stereo endoscopic images. Unlike prior work, our fully feed-forward pipeline eliminates the need for offline preprocessing or per-scene optimization, significantly enhancing its potential for practical application in surgical environments.

The proposed framework integrates a DilatedResNet-based depth estimation module with a Gaussian attribute predictor to jointly estimate scene geometry and appearance parameters. Experiments on the ENDONERF and SCARED datasets demonstrate that Endo-E2E-GS consistently achieves higher rendering fidelity than state-of-the-art

baselines, while avoiding the complexity of scene-specific initialization and maintaining sub-real-time but moderate computational cost. Ablation studies further validate the contributions of depth pretraining and hierarchical feature encoding, both of which significantly improve reconstruction quality and stability.

Despite its promising results, key limitations in real-time performance and temporal consistency remain. Future work will focus on improving computational efficiency and incorporating temporal models to bridge the gap between our current framework and a clinically deployable system. Overcoming these challenges will be crucial for unlocking the full potential of real-time 3D reconstruction in the operating room.

CRediT authorship contribution statement

Xiongzhi Wang: Investigation, Methodology, Validation, Writing – original draft. **Boyu Yang:** Data curation, Writing – review & editing. **Min Wei:** Data curation, Writing – review & editing. **Yu Chen:** Data curation, Writing – review & editing. **Jingang Zhang:** Supervision, Writing – review & editing. **Yunfeng Nie:** Conceptualization, Methodology, Writing – original draft, Review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The source code is available at: https://github.com/Intelligent-Imaging-Center/Endo-E2E-GS. We used two publicly available stereo endoscopic datasets—ENDONERF and SCARED—as detailed in the manuscript.

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